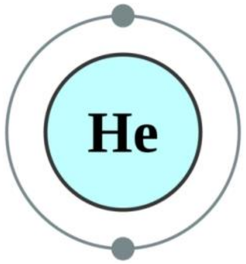
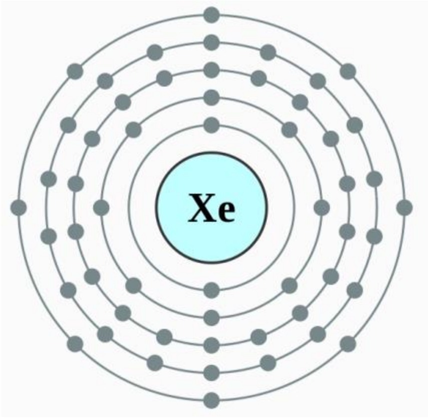
Ion/Molecule Reactions in MS

Permanent dipole and polarizability of atoms and molecules

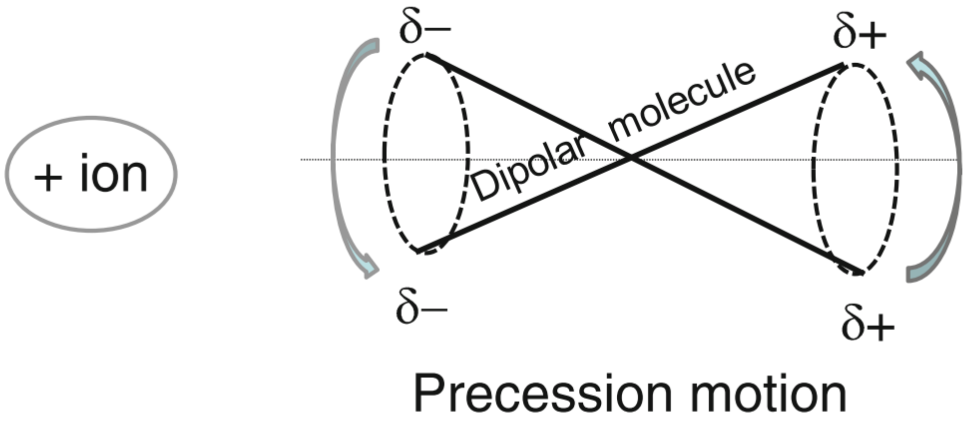


Atom, molecule	Permanent dipole, μ(Debye)	Polarizability, α (A^3)
Rare gas atoms		
He	0	0.21
Ne	0	0.41
Ar	0	1.64
Kr	0	2.48
Xe	0	4.02
Diatomic		
H_2	0	0.81
D_2	0	0.8
O_2	0	1.57
N_2	0	1.74
CO	0.1	1.94
NO	0.16	1.7
HCl	1.03	2.58
HBr	0.79	3.49
Triatomic molecule		
CO_2	0	2.59
CS_2	0	8.08
H_2O	1.84	1.45
N_2O	0.17	2.92
H_2S	1.02	3.61
SO_2	1.62	3.78
Polyatomic molecule		
CH_4	0	2.56
CCl_4	0	10.24
C_2H_4	0	4.1
C_2H_6	0	4.39
n-C_4H_{10}	0	8
i-C_4H_{10}	0	8
C_6H_6	0	9.99
NH_3	1.47	2.16
H_2CO	2.31	2.81
CH_3OH	1.7	3.25
$(CH_3)_2CO$	2.85	6.11



The measure of “softness” of an electron cloud is quantitatively given by the “polarizability” (α).

离子靠近极性分子的过程



Types of Ion/molecule reactions

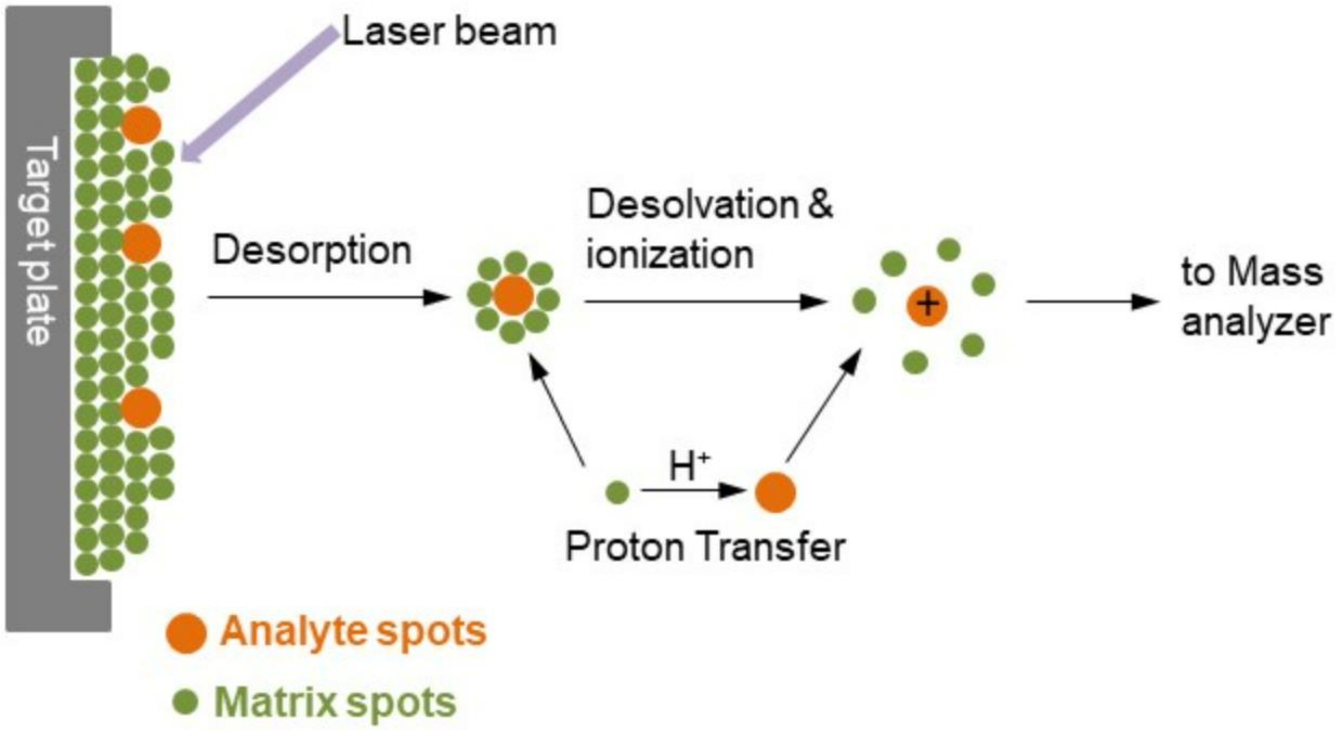
- Electron Transfer Reactions
- Proton Transfer Reactions
- Hydride Transfer Reactions
- Clustering Reactions
- Negative Ion Formation
- SN2 displacement reactions

Electron Transfer Reactions

$$A^+ + B \longrightarrow A + B^+$$

Proton Transfer Reactions

MALDI（Matrix-Assisted Laser Desorption/Ionization，基质辅助激光解吸电离）



https://www.creative-proteomics.com/

1. 样品制备

选择基质：选择适合分析物的基质，常用基质包括α-氰基-4-羟基肉桂酸（CHCA）和二羟基苯甲酸（DHB）。

混合样品与基质：将样品与基质溶液混合，通常比例为1:1000到1:10000。

点样：将混合液滴在MALDI靶板上，待溶剂挥发后形成共结晶。

2. 离子源准备

靶板装载：将点样后的靶板放入MALDI离子源。

真空系统：启动真空系统，确保离子源内部达到所需真空度。

3. 激光照射

激光参数设置：根据样品特性调整激光波长（通常为337 nm）和能量。

激光照射：激光脉冲照射样品-基质共结晶区域，使基质吸收能量并解吸样品分子。

4. 离子化过程

能量转移：基质吸收激光能量并转移给样品分子，使其离子化。

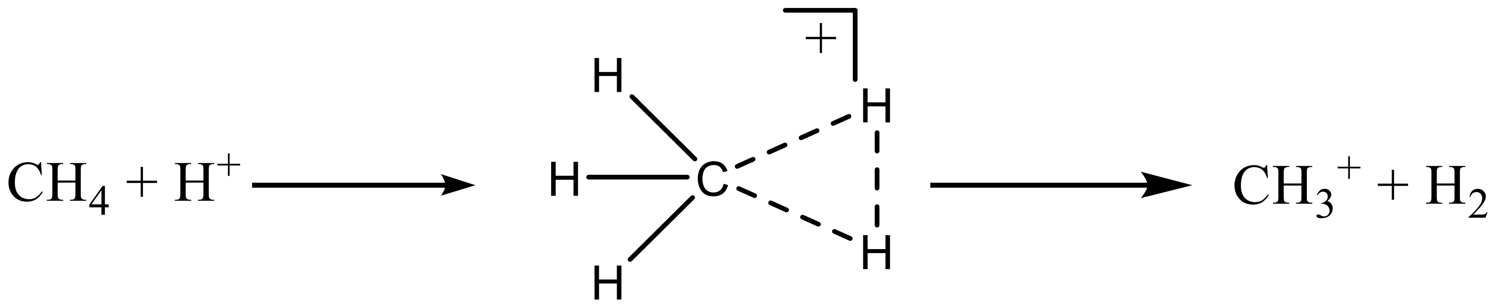
5. 离子生成

样品分子形成正离子或负离子，通常为单电荷离子。

化学电离（CI）的质子转移：

$$CH_5^+ + NH_3 \rightleftharpoons CH_4 + NH_4^+$$

Hydride Transfer Reactions



Negative Ion Formation

Electron capture:

$$AB + e^- \underset{k_b}{\overset{k_c}{\rightleftharpoons}} AB^{-*}$$

$$AB^{-*} \longrightarrow A + B^-$$

$$AB + e^- \underset{k_b}{\overset{k_c}{\rightleftharpoons}} AB^{-*} + M \longrightarrow AB^- + M \text{ (M: third body)}$$

Ion pair formation:

$$AB + e^- \longrightarrow A^+ + B^- + e^-$$

Charge transfer:

$$AB + X \longrightarrow AB^- + X^+$$

Electron capture reactions frequently used for CI

$$O_2 + e^- \longrightarrow O_2^- \longrightarrow O^- + O$$

$$SF_6 + e^- \longrightarrow SF_6^- \longrightarrow SF_5^- + F$$

$$CCl_4 + e^- \longrightarrow Cl^- + CCl_3$$

Electron affinity (eV)

Chemical species	Electron affinity
H	0.75
O	1.46
F	3.4
Cl	3.62
Br	3.37
I	3.07
NO	0.024
NO_2	2.43
NO_3	3.7
N_2O	0.22
O_2	0.5
O_3	2.05
OH	1.83
SCN	2.17
SF_6	0.6
SF_5	2.8
7,7',8,8'-tetracyanoquinodimethane (TCNQ)	2.83
Nitrobenzene	0.98
Tetracene	0.88
Azulene	0.66
Anthracene	0.55
Naphthalene	0.14

A typical example of SN2 reaction is the halide ion displacement reaction:

$$Cl^{-} + CH_3Br = ClCH_3 + Br^{-}$$